

Stacked Polyphase Bridge Converters for High-Voltage Electric Traction Drives: A Review of Topology, Modulation, Control, Semiconductor Scaling, and Research Challenges

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Abstract—Higher dc-link voltage is increasingly attractive in electric traction, but it raises semiconductor blocking-voltage, switching-transient, insulation, and electromagnetic-compatibility demands. The stacked polyphase bridge (SPB) addresses the semiconductor-voltage problem by connecting three-phase two-level bridges in series on the dc side, each supplying an electrically separated winding group — a voltage-partitioning approach that accesses lower-voltage semiconductor classes and supports modular converter-machine integration, at the cost of added winding, dc-link, sensing, gate-drive, and insulation-coordination overhead. Although SPB has been studied for over a decade, the literature remains scattered across topology, modulation, common-mode voltage, dc-link balancing, multiphase-machine design, semiconductor technology, and fault-tolerant control, without a common system-boundary evaluation. This review consolidates SPB’s evolution, operating principle, and machine-integration options (aligned, spatially shifted, and physically segmented winding groups); derives the double Fourier series basis for its interleaved-carrier dc-link current ripple; quantifies common-mode-voltage cancellation, switching-delay-induced residual spikes, and the resulting bearing current; analyzes dc-link voltage-balancing stability from the exact nonlinear cell model across centralized, common-duty-ratio, and distributed control; and examines semiconductor scaling and post-fault operation using reported experimental fault data. A cell-count-to-voltage-class mapping with a commercial-device screening approach is developed for a 1200-V dc link, showing that lower device-voltage classes can yield nonlinear semiconductor-level gains while converter and machine hardware scale roughly with cell count, so the preferred cell count is mission-dependent rather than fixed. SPB should therefore be assessed as a modular converter-machine architecture whose competitiveness depends on whether voltage partitioning yields a system-level benefit sufficient to justify its added integration and control complexity.

Index Terms—Electric vehicles, integrated motor drives, modular converters, multiphase machines, stacked polyphase bridge (SPB) converters.

I. INTRODUCTION

Electrification is pushing traction-drive propulsion and fast-charging power toward several hundred kilowatts, while system-level performance requirements continue to rise [1]. Increasing the dc-link voltage has emerged as a primary design lever for meeting both demands. On the propulsion side, a higher dc-link voltage improves efficiency and power

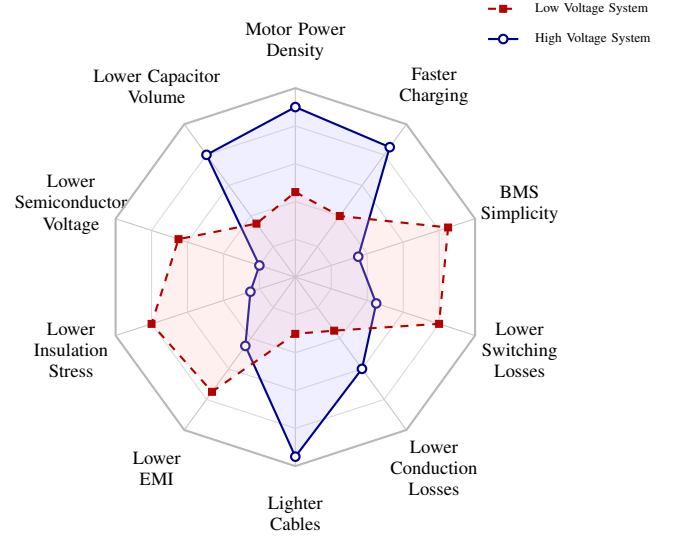


Fig. 1. Conceptual illustration of the qualitative system-level tradeoffs between a low-voltage and a high-voltage traction architecture.

density [2]. On the battery and charging side, it reduces the current required for a given power level, lowering cable conduction loss and conductor cross-section while enabling higher charging power under a given current limit [3].

This trend, however, is not without consequence at the converter level. A higher dc-link voltage also brings more severe insulation-coordination and electromagnetic-compatibility (EMC) demands, together with added thermal and reliability stress on the power stage [2], [3]. Fig. 1 summarizes this qualitative tradeoff between a conventional low-voltage and a higher-voltage traction architecture. Raising the dc-link voltage therefore does not amount to a simple rescaling of an existing inverter design; it reshapes the architecture-level design space of the complete electric drive.

A. Literature and Industry Trends Toward Higher DC-Link Voltage

The trend motivated above is not merely conceptual: it is already established in academic prototypes, industrial inverter platforms, and production vehicles, as summarized in Table I. Reported academic demonstrators range from 400-V to 800-V and 1.2-kV converters, with still higher levels investigated for high-power passenger and heavy-duty applications [3], while

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TABLE I

REPRESENTATIVE VOLTAGE AND POWER LEVELS IN PUBLISHED TRACTION-INVERTER STUDIES, INDUSTRIAL AND REFERENCE PLATFORMS, AND PRODUCTION BATTERY-ELECTRIC VEHICLES.

Published studies / demonstrators	Industrial / reference platforms	Production vehicles
Bertelshofer et al.: 400 V, 150 kW [6]	Bosch Gen. 4: 400/800 V, 250/350 kW [7]	BMW i4 M50: 400 V, 400 kW [8]
Taha et al.: 800 V, 100 kW [9]	Wolfspeed reference design: 800 V, 300 kW [10]	Hyundai IONIQ 5 N: 800 V, 478 kW [11]
Al-Hmoud et al.: 1 kV, 200 kW [12]	Infineon eCAV platform: 800 V, 250 kW [13]	Volvo EX90 Performance: 800 V, 500 kW [14]
Wang et al.: 1.2 kV, 100 kW [15]	AVL PI850e: 800 V, 250 kW [16]	Porsche Taycan Turbo S: 800 V, 700 kW [17]
Ayaz et al.: 1.2 kV, 250 kW [18]	Punch IV5: 800 V, 600 kW [19]	Lucid Air Dream Performance: 900 V, 828 kW [20]

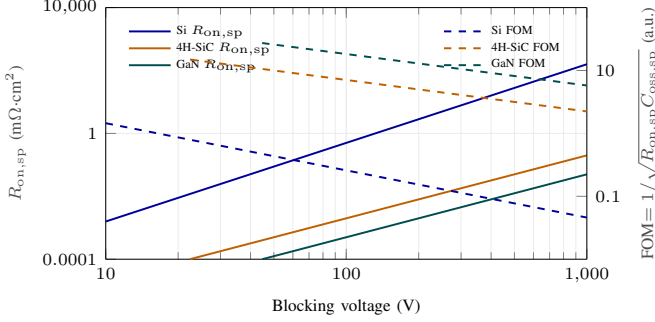


Fig. 2. Ideal one-dimensional unipolar specific on-state resistance $R_{on,sp}$ (left axis, solid) and the resulting high-frequency figure of merit $FOM = 1/\sqrt{R_{on,sp}C_{oss,sp}}$ (right axis, dashed, arbitrary units) versus blocking voltage for Si, 4H-SiC, and GaN.

industrial platforms increasingly target 800-V operation as a baseline and production vehicles show a comparable shift from established 400-V architectures to a growing number of 800-V and higher-voltage platforms [4], [5]. This industry-wide direction, rather than an isolated research interest, motivates the semiconductor-level analysis that follows since the rising dc-link voltages translate directly into a semiconductor-level bottleneck.

B. Semiconductor Blocking-Voltage Tradeoff

In a conventional two-level voltage-source inverter, each active switch must block a voltage directly tied to the complete dc-link voltage. In unipolar power devices, a higher blocking-voltage rating imposes a fundamental tradeoff against both specific on-state resistance and switching performance, commonly captured by the figure of merit, as shown in Fig. 2 [21], [22]. Therefore, a lower-voltage device consistently achieves a higher FOM than a higher-voltage device of the same material.

Wide-bandgap (WBG) semiconductors shift this FOM favorably: SiC metal-oxide-semiconductor field-effect transistors (MOSFETs) now dominate high-voltage automotive drives, while GaN is particularly attractive at lower voltage owing to its fast switching behavior and superior FOM [21], [23]–[25]. Yet switching material does not remove the underlying coupling between dc-link voltage and device voltage class: irrespective of technology, a higher dc-link voltage still forces a higher-voltage, higher- $R_{on,sp}$ device and higher switching losses. Converter architectures that partition the dc-link voltage among multiple switching units, so each device blocks only a fraction of the total voltage, therefore become increasingly relevant as system voltage rises.

C. Multilevel Converters and Their Limitations

Multilevel converters constitute an established solution to this voltage-partitioning problem. NPC, T-type, flying-capacitor, and cascaded architectures all distribute the dc-link voltage among additional semiconductor positions or capacitor-voltage levels, reducing individual device voltage stress [26]–[29].

This voltage partitioning, however, comes at the cost of additional semiconductor devices, passive components, and gate drivers – each an additional potential failure point – without, by itself, yielding a modular converter–machine system: because the added semiconductor positions continue to feed a single, electrically common three-phase winding set, a fault at one position cannot generally be isolated and bypassed without disabling the drive. Fault tolerance, power redistribution, and post-fault reconfiguration can be implemented in conventional multilevel drives, but typically require additional redundancy, dedicated isolation paths, and specialized control layered on top of the base topology rather than following from it. A larger semiconductor count therefore does not yield a commensurate increase in fault tolerance, motivating an alternative form of voltage partitioning.

D. Stacked Polyphase Bridge

Unlike other multilevel converters, which distribute the dc-link voltage among additional semiconductor positions within a single converter, the stacked polyphase bridge (SPB) distributes the high-voltage conversion function among N repeated converter units. Each unit is a conventional three-phase two-level bridge connected in series on the dc side, supplying its own electrically separated three-phase winding group of the same machine, as illustrated in Fig. 3.

Under balanced operation, each cell’s local dc-link voltage $V_{dc,k} \approx V_{dc}/N$, so the semiconductor voltage rating can be selected on the local cell voltage rather than the complete traction dc-link voltage – addressing the semiconductor bottleneck much as multilevel converters do. Because each cell also drives its own separated winding group, however, SPB additionally has an inherently modular structure that multilevel converters lack: a faulted cell can, in principle, be isolated and bypassed while the remaining cells continue to drive the machine at reduced power. It is this combination of semiconductor voltage reduction and converter–machine modularity that distinguishes SPB from conventional multilevel and cascaded architectures and motivates its consideration as a candidate topology for next-generation traction drives.

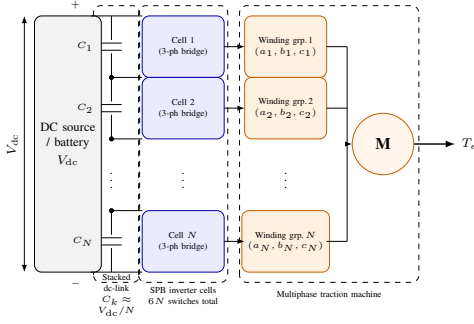


Fig. 3. Conceptual architecture of an N -cell SPB traction drive. The dc source V_{dc} feeds N series-connected local dc-link capacitors, each nominally V_{dc}/N , that individually supply N series-connected three-phase two-level bridge cells ($6N$ switches total); each cell drives one electrically separated three-phase winding group of the same machine, and the combined winding groups produce the electromagnetic torque T_e .

SPB was introduced by Norrga *et al.* for compact EV and HEV drives [30]; its subsequent development toward higher-fidelity control, balancing, and reconfiguration is traced in detail in Section II.

Realizing this potential, however, introduces its own design challenges: component count, dc-link dynamics, machine design, common-mode voltage and insulation coordination, and fault handling all become more tightly coupled, as examined in the sections that follow, with thermal management identified as an open research direction (Section X).

These questions have grown more relevant as neighboring research areas have matured: multiphase machines provide additional post-fault degrees of freedom, integrated modular motor drives show how repeated power-electronic units can be physically associated with individual machine sections, wide-bandgap semiconductors amplify performance differences among voltage classes, and distributed control coordinates modular converter units [31]–[36] – together motivating SPB’s assessment as a complete converter–machine architecture rather than only a series-connected topology.

Despite more than a decade of research on SPB, however, no study has yet brought these threads together to systematically assess its feasibility for electric-vehicle traction applications. The available literature remains distributed across converter topology, modulation, dc-link stability, multiphase-machine design, integrated motor drives, semiconductor technology, thermal management, distributed control, and fault-tolerant operation. Although these topics are strongly coupled in an SPB traction drive, they are rarely evaluated within a common system boundary. Moreover, most reported SPB experimental demonstrations remain validated only at laboratory scale (Section II), even as the enabling semiconductor, machine, cooling, and packaging technologies continue to advance rapidly. This gap between a well-motivated architectural concept and a system-level feasibility assessment is the gap that the present paper addresses.

Accordingly, this paper provides a system-level review of SPB traction-drive technology aimed at closing that gap. The remainder of the paper is organized as follows. First, the historical development, operating principle, and modulation

of SPB are consolidated. Second, the interaction between SPB cell count and the electric machine is examined together with relevant multiphase-machine and integrated modular motor-drive research, including the physical-integration and insulation-coordination overhead introduced by series-connected converter cells. Third, the series-stack origin of common-mode voltage in SPB, its physical consequences, and mitigation options are examined, followed by dc-link dynamics and balancing. Fourth, semiconductor voltage scaling, SiC and GaN device selection, loss scaling, and fault handling and reconfiguration are considered as coupled architecture-level design problems. Finally, the research gaps and requirements for high-power automotive validation are identified, including high-power EMC validation and the quantitative comparison against conventional two-level, multilevel, and cascaded alternatives that remain open for future work.

II. EVOLUTION OF THE STACKED POLYPHASE BRIDGE CONCEPT

This section traces how SPB’s converter–machine modularity, together with its associated control and modulation methods, has developed over more than a decade of research, from an initial concept to an increasingly reconfigurable and actively controlled converter–machine system.

SPB emerged from a broader development toward modular converter–machine systems. Earlier modular and integrated modular motor drives (IMMDs) established the building blocks SPB later drew on, repeated converter-winding modules, machine segmentation, power sharing, and fault-tolerant operation [31], [37], and series-connected IMMD modules further showed that the total dc voltage could be shared among inverter segments to enable lower-voltage devices [32], [38]. A 2015 IEEE TIA study by Wang, Li, and Han is especially relevant to SPB’s later work, combining series-connected GaN IMMD hardware with active cell-voltage balancing and interleaved switching [39].

Independent of this line of work, other series-connected modular motor-drive topologies were proposed with the same underlying dc-voltage-sharing principle [40], [41], indicating that the concept was not unique to SPB but part of a broader convergence toward series-connected modularity. A related KTH comparative study evaluated SPB alongside other integrated modular converter candidates for traction drives, the parallel-connected polyphase bridge converter (PPB-I, a conventional multiphase converter with phase-leg-level modularity, and PPB-II, several interleaved two-level three-phase converters each with its own dc capacitor) and the modular high-frequency (MHF) converter, against a conventional two-level three-phase drive as a baseline in terms of machine design, losses, capacitor energy storage, cost, and redundancy, confirming that SPB is one of several modular-converter topology candidates studied for this application [42].

Building on this foundation, the SPB topology itself was introduced in 2013 [30], followed shortly thereafter by dedicated control and modulation work [43]. Related series-connected multiphase drives demonstrated that machine-side degrees of freedom could be used to regulate internal dc-link voltage

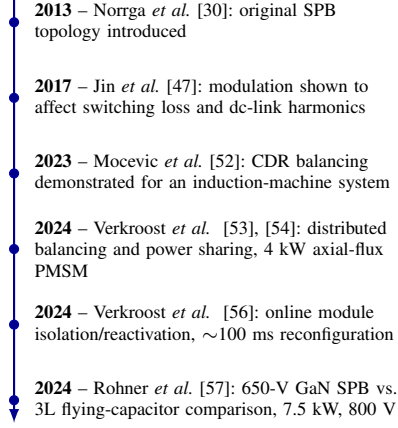


Fig. 4. Chronological milestones in SPB research, from the original topology through modulation, distributed balancing, reconfiguration, and wide-bandgap integration.

sharing [44], while Nikouie *et al.* formally established the SPB dc-link stability problem and the corresponding balancing-controller requirements [45], [46]. This period established the topology’s basic feasibility and identified dc-link voltage balancing, rather than semiconductor voltage rating alone, as its first-order control challenge.

The next stage of development shifted the focus toward modulation, scalable control, and converter–machine co-design: modulation was shown to directly influence semiconductor loss and dc-link harmonic content [47], distributed control offered an alternative to centralized balancing [48], and interleaving and phase-count studies tied converter design to machine architecture [34], [49]–[51]. Collectively, this reframed SPB from a series-connected converter topology into a converter–machine co-design problem in which modulation, control, and machine parameters jointly determine dc-link behavior.

More recent work has moved from repeated but statically configured hardware toward actively reconfigurable modular systems. Common-duty-ratio operation, decentralized voltage balancing, and multi-agent power distribution have reduced the dependence on centralized control [52]–[55]. Online module isolation and reintroduction have further demonstrated dynamic dc-bus reconfiguration, removing a faulted cell without interrupting torque production [56]. In parallel, modern GaN devices have renewed interest in SPB as a route to lower semiconductor voltage classes in integrated drives [57].

Fig. 4 places representative milestones from across these stages on a chronological timeline of the topology’s development, together with the validation level, reported scale, and the specific implication each result carries for the system-level review developed in this paper.

As Fig. 4 indicates, this progression has been validated primarily at laboratory scale, typically at or below several kilowatts, leaving open whether the demonstrated control, balancing, and reconfiguration principles extend to the automotive and heavy-duty power levels summarized in Table I – a gap revisited throughout the remainder of this paper.

III. OPERATING PRINCIPLE AND TOPOLOGICAL CHARACTERISTICS

This section formalizes the operating principle underlying SPB’s two coupled properties – reduced semiconductor voltage stress and converter–machine modularity – and establishes the notation and voltage/power relations used throughout the remainder of the paper.

A. Basic Structure and Voltage Partitioning

An N -cell SPB consists of N conventional three-phase two-level voltage-source inverter cells connected in series on the dc side, with each bridge supplying one electrically separated three-phase winding group of the same machine [30], [43], [47]. The machine therefore presents N ac ports, while the converter cells share a single series dc path. This structure is the physical implementation of the voltage-partitioning principle introduced in Section I-B.

The dc-link voltage satisfies

$$V_{dc} = \sum_{k=1}^N V_{dc,k}, \quad V_{dc,k} \approx \frac{V_{dc}}{N} \quad (1)$$

under balanced operation. A basic implementation contains $6N$ active switches and N local dc-link capacitors. Increasing N therefore reduces the nominal cell voltage, and hence the required semiconductor voltage class, but simultaneously increases the number of switches, gate drivers, sensors, capacitors, and machine-winding terminals that must be realized and managed. This tradeoff between semiconductor voltage rating and component count is quantified further in Section VIII and is revisited from a machine-design perspective below. Fig. 5 shows this structure at switch level for two representative cells, making explicit the $6N$ -switch bridge realization, the local capacitors C_k , and the notation used throughout the remainder of this paper.

B. Series DC Connection and Power Flow

The defining structural difference between SPB and parallel modular drives is the series dc connection. Neglecting auxiliary and bypass paths, the cell dc currents are constrained to be equal,

$$i_{dc,1} = i_{dc,2} = \dots = i_{dc,N} = i_{dc}, \quad (2)$$

so that the dc-side power entering cell k is

$$p_{dc,k} = V_{dc,k} i_{dc}. \quad (3)$$

The total machine power, by contrast, is the sum of the powers delivered to the individual winding groups,

$$P_{ac} = \sum_{k=1}^N P_k. \quad (4)$$

Under symmetrical operation $P_k \approx P_{ac}/N$, but equal power sharing is not a fundamental requirement of the topology. The multi-three-phase machine provides additional current and power degrees of freedom that allow power to be redistributed

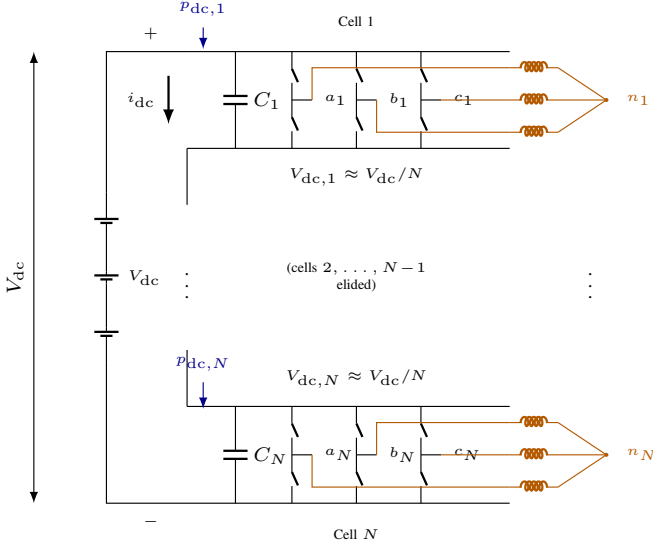


Fig. 5. Switch-level realization of the SPB dc-ac stage for two representative cells (Cell 1 and Cell N ; cells $2, \dots, N-1$ elided). Each cell is a conventional two-level three-phase bridge (six switches per cell, $6N$ total) fed from its own local dc-link capacitor C_k , with leg midpoints a_k, b_k, c_k feeding a star-connected winding group k at its own electrically separate neutral point n_k (only n_1 and n_N shown; the star points are not interconnected). The common series current i_{dc} (2) flows through every cell, giving each cell the dc-side power $p_{dc,k} = V_{dc,k} i_{dc}$ (3).

among winding groups while the required total torque is maintained [44], [54]. This redistribution capability is the physical basis for both dc-link voltage balancing and several of the post-fault operating strategies discussed in later sections.

The combination of a common dc current with independently controllable ac-side powers also explains why the local capacitor voltages must be treated as internal dynamic states rather than fixed design parameters. If one winding group processes more power than another, the corresponding capacitor energies can diverge even though all cells carry the same dc current [45], [46]. Managing this divergence is precisely the dc-link stability and balancing problem identified in Section II as the topology's first-order control challenge.

C. SPB as an Input-Series, Multi-Three-Phase-Output Drive

These relations place SPB in a distinct category relative to the conventional multilevel converters discussed in Section I, which partition dc-link voltage internally while retaining a conventional three-phase machine. In SPB, by contrast, each converter cell remains a standard two-level bridge, and it is the machine itself that provides the multiple ac ports required to transfer power out of the series dc stack. SPB can therefore be characterized as an *input-series, multi-three-phase-output* drive, a description that makes explicit why voltage partitioning in SPB cannot be separated from machine design.

D. Cell Count as a Converter-Machine Design Variable

Cell count is consequently a converter-machine design variable rather than a semiconductor variable alone. As established above, increasing N also reduces the voltage available to each winding group and alters the number of winding sets, terminal

count, electromagnetic coupling, current-control degrees of freedom, and available reconfiguration states. Bringezu and Biela demonstrated this interaction directly by comparing SPB systems based on 3-, 6-, 9-, and 12-phase PMSMs [34].

A related and often overlooked point is that machine phase current does not generally scale as $1/N$. The required phase current instead depends on the number of turns, flux linkage, winding connection, torque constant, and machine geometry. A many-cell architecture may therefore reduce the semiconductor voltage class while still requiring high-current devices or parallel chips at the device level. This coupling between cell count, voltage class, and current rating is particularly important when the comparison is extended from high-voltage SiC implementations toward lower-voltage GaN implementations, as considered in Section I-B.

IV. ELECTRIC-MACHINE REQUIREMENTS AND INTEGRATED IMPLEMENTATION

SPB is, by construction, an input-series, multi-three-phase-output drive (Section III): an N -cell design requires N electrically separated three-phase winding groups, each connected to one converter cell, coupling semiconductor voltage scaling directly to winding segmentation, phase count, terminal count, electromagnetic coupling, insulation, and physical machine integration.

A. Winding Architecture and Electromagnetic Coupling

Although the winding groups are electrically separated at their terminals, they share the same magnetic circuit, rotor, air gap, and shaft. Their electromagnetic coupling depends on winding distribution, slot/pole combination, spatial displacement, and machine geometry [34], [44], [54], [58]. Therefore, an SPB-compatible machine cannot generally be obtained simply by dividing a conventional three-phase winding into multiple groups without considering the resulting coupling and harmonic behavior.

The three-phase groups may be approximately aligned, deliberately spatially shifted, or implemented as physically segmented stator modules. Spatial displacement affects mutual coupling, torque-ripple components, and the additional current subspaces available to the controller, among other effects [59]–[61]. These additional subspaces can introduce non-torque-producing currents, but they can also be exploited for harmonic suppression, differential power sharing, and fault-tolerant operation [54], [62]. Split winding architectures show a similar tradeoff: they can reduce ripple and provide additional current/voltage degrees of freedom, but they do so by increasing winding terminals and coupling complexity [63]. These three winding-group arrangements are compared schematically in Fig. 6.

For the k th PMSM winding group,

$$\begin{aligned} v_{d,k} &= R_{s,k} i_{d,k} - \omega_e L_{q,k} i_{q,k}, \\ v_{q,k} &= R_{s,k} i_{q,k} + \omega_e (L_{d,k} i_{d,k} + \psi_{f,k}), \end{aligned} \quad (5)$$

and the required voltage must satisfy

$$\sqrt{v_{d,k}^2 + v_{q,k}^2} \leq k_{\text{mod}} V_{dc,k} \approx k_{\text{mod}} \frac{V_{dc}}{N}. \quad (6)$$

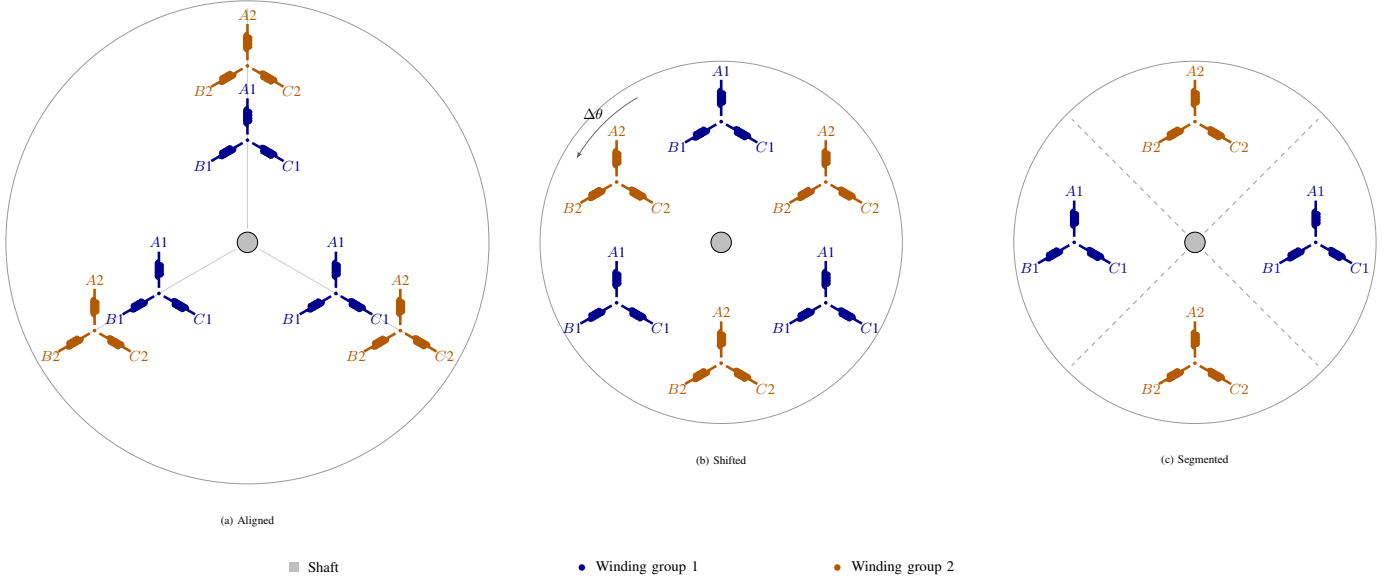


Fig. 6. Schematic comparison of the winding-group spatial arrangements discussed in the text, drawn as star-connected coil groups (each leg a winding phase, small dot the group's local neutral point) inside the stator bore, with the shaft centerline marked for physical context (small grey disk): (a) approximately aligned groups – at every one of three representative slots around the bore, group 1 (inner layer) and group 2 (outer layer) occupy the same angular position at different radial depth; (b) deliberately spatially shifted groups – both groups sit at the same radial layer, but group 2 is displaced from group 1 by a fixed angle $\Delta\theta = 60^\circ$ at every slot, so the six coil clusters interleave evenly around the bore; and (c) physically segmented stator modules – the same bore split by four mechanical/magnetic boundaries (dashed radial lines) into four modules that alternate between group 1 and group 2, so each group occupies two diametrically opposite, physically separate segments rather than a single contiguous arc. The arrangement affects mutual coupling, MMF harmonics, and the additional current subspaces available to the controller.

Increasing the number of cells therefore reduces the voltage available to each winding group and requires coordinated selection of winding turns, inductance, and field-weakening capability, among other parameters. This coupling has been demonstrated in optimized multiphase SPB-PMSM systems [34]. Recent multiphase traction-machine studies similarly emphasize coordinated design of winding voltage, harmonic subspaces, and modulation [64], [65].

Phase current does not generally scale with $1/N$ (Section III); this is particularly important for low-voltage WBG implementations, where reduced blocking voltage may still be accompanied by high converter-cell current.

B. From Multiphase Machine to Integrated Modular Drive

Physical integration can convert part of the additional electrical complexity introduced by a large number of winding groups into a packaging advantage. With a remote inverter, additional winding groups increase phase-cable and connector count. In an IMMD, each converter bridge can instead be positioned close to its corresponding winding group, reducing external cabling, shortening high-frequency interconnections, distributing heat sources, and aligning converter modules with stator segments.

This design philosophy predates SPB and has been extensively studied in integrated and modular motor drives [31], [32], [35], [38], [39], [66]–[78], summarized in Table II. The overall trend runs from early cable-reduction and compact-packaging concepts through jointly designed fault-tolerant converter/winding/thermal architectures and series-connected GaN IMMDs to recent WBG-based axial-flux IMMDs demon-

strating stator-mounted converter modules and scalability to MW-class and EV-traction implementations.

C. Physical Integration Options

The electrically separated winding groups required by SPB do not impose a unique converter placement on a radial-flux machine. Chowdhury *et al.* [79], consistent with the broader integrated-motor-drive survey of Abebe *et al.* [80], classify motor/inverter physical integration into four categories, distinguished by whether the converter mounts radially or axially and whether it interfaces with a separate housing/end-cap structure or directly with the stator core itself; this taxonomy provides a convenient framework for placing SPB converter cells around a radial-flux machine, as illustrated in Fig. 7. (Axial-flux stator-mounted and polygon-housing integration, an alternative to the radial-flux options considered here, is discussed above in the context of the Mohamed *et al.* developments [74]–[76] and Table II.)

In a *radial housing mount* [Fig. 7(a)], converter modules are mounted on the outer motor housing, physically and thermally set back from the stator by the housing structure. In a *radial stator mount* [Fig. 7(b)], modules are instead mounted flush against the stator's outer surface or cooling jacket, removing that separation at the cost of requiring a mounting-compatible stator periphery. Along the axial direction, an *axial endplate mount* [Fig. 7(c)] places modules on a separate end-cap beyond the machine's active length, whereas an *axial stator mount* [Fig. 7(d)] places them directly against the end face of the stator lamination stack, without an intervening cap structure. Dedicated finite-element thermal comparisons of these mount

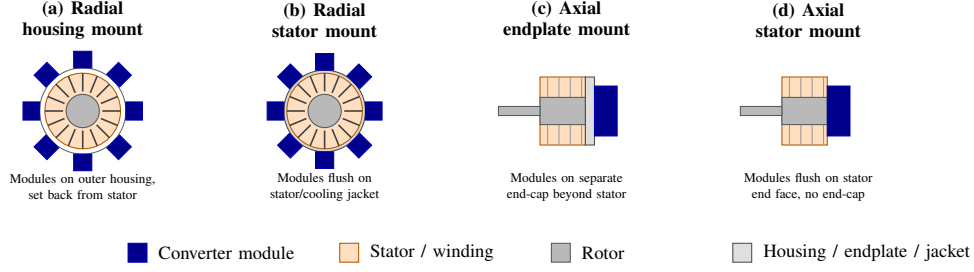


Fig. 7. Physical integration options for SPB-compatible converter modules on a radial-flux machine, following the four-category taxonomy of [79]: (a) radial housing mount, (b) radial stator mount, (c) axial endplate mount, and (d) axial stator mount, each shown in whichever radial or axial cross-section most clearly conveys its mounting geometry.

TABLE II
REPRESENTATIVE INTEGRATED AND MODULAR MOTOR-DRIVE
DEVELOPMENTS RELEVANT TO SPB

Reference	Integration theme	Relevance to SPB
Shakweh <i>et al.</i> [66]	Integrated motor/inverter architecture	Practical benefits and high-power integration barriers
Wheeler/Pickering <i>et al.</i> [67], [68]	Converter and thermal design inside motor envelope	Importance of coupled packaging and cooling
Brown <i>et al.</i> [69]	Converter mounted at modular stator pole	Repeated converter-winding hardware as an architecture
Wolmarans <i>et al.</i> [70]	Fault-tolerant converter, machine, and thermal design	Links modularity with fault tolerance and cooling
Choi <i>et al.</i> [31]	Modular PM machine and high-temperature electronics	Traction-oriented precursor to converter-machine modularization
Wang <i>et al.</i> [32], [38]	Low-voltage GaN, dc-voltage sharing, and interleaving	Precursor to WBG-enabled SPB voltage partitioning
Uğur and Keysan [72], [73]	Multi-physics converter-machine optimization	Connects GaN selection, capacitance, cooling, machine design
Mohamed <i>et al.</i> [74]–[76]	Stator-mounted modules, polygonal housing, integrated dc link	Directly relevant to physical integration of repeated cells
Wang <i>et al.</i> [77]	Repeated machine and converter submodules	Scalability of modular integration to high power
Chen <i>et al.</i> [78]	WBG converter, machine, passive, and cooling co-design	Recent traction-oriented integrated-drive context
Deng <i>et al.</i> [35]	Recent IMD technologies and research directions	Context for packaging, cooling, reliability, manufacturing

options support this picture: an FEA study of radial-housing, radial-stator, and endplate-mounted converters found the modularized endplate arrangement most effective at limiting peak temperature and improving temperature distribution across the assembly [81].

Modern WBG drives therefore provide several machine and packaging concepts that are compatible with SPB, including segmented windings, distributed converter cells, shared cooling, and integrated dc-link structures. These developments do not establish an intrinsic SPB advantage, but they demonstrate the availability of enabling technologies required for practical implementation.

D. Integration Is Not Free of Complexity

Integration does not remove complexity; it relocates it. Converter modules mounted on or near the machine are exposed to elevated temperature, vibration, and sealing/coolant requirements, among other stresses, and to electromagnetic coupling between gate-drive, power, and machine conductors. Shared machine-converter cooling can also increase thermal interaction between the stator and power electronics.

SPB introduces an additional insulation challenge because its converter cells and associated winding groups occupy different absolute potentials along the series dc stack. Gate-driver supplies, sensors, and mechanical mounting, among other subsystems, must therefore preserve the required electrical isolation.

Consequently, practical SPB integration requires coordinated electrical, thermal, and mechanical design, complementing the semiconductor- and system-level tradeoffs discussed in the following sections.

V. MODULATION STRATEGIES

Each SPB cell is electrically simple, a conventional two-level bridge (Sections III and IV), while the resulting control problem is not: cell dc current is shared, cell dc voltage is a dynamic state, and the winding groups are electromagnetically coupled. Modulation is the layer at which these converter- and machine-side constraints are jointly realized in real time.

Because each SPB cell is a conventional three-phase two-level bridge, standard SPWM, SVM, and DPWM methods can be applied locally without modification. The distinctive feature of SPB is that modulation can additionally be coordinated *between* cells through carrier phase, zero-sequence injection, switching sequence, and clamping intervals [43], [47]. These inter-cell degrees of freedom affect dc-link ripple, machine harmonics, semiconductor switching loss, common-mode voltage (CMV), and the power exchanged by the individual winding groups, and they have no counterpart in a single conventional three-phase inverter.

A. Carrier Coordination and Harmonic Cancellation

For carrier-based PWM, uniform interleaving sets

$$\phi_{c,k} = \phi_{c,1} + (k-1)\Delta\phi_c, \quad \Delta\phi_c = \frac{2\pi}{N}. \quad (7)$$

A switching harmonic of similar amplitude in all cells then combines approximately as

$$I_{h,\Sigma} = I_h \sum_{k=0}^{N-1} e^{j h k \Delta \phi_c}, \quad (8)$$

so that an appropriate phase displacement can suppress selected switching-frequency components, as illustrated for the $N = 3$ case discussed throughout this subsection. SPB-specific studies have shown that inter-cell modulation affects semiconductor losses and dc-link-current harmonics [47]; related modular and multiphase drives further demonstrate simultaneous reduction of dc-link and machine current ripple [51], [82], [83].

Equation (8) treats each cell's ripple contribution as a single tone at the switching frequency; a rigorous double-Fourier-series analysis of the actual switched converter current gives the exact survival condition instead of this single-tone approximation. Writing the switching function of phase leg x in cell k in the usual double-Fourier form, with carrier index m (at carrier frequency $\omega_{c,k}$, phase $\theta_{c,k}$) and sideband index n (at fundamental frequency $\omega_{0,k}$, phase $\theta_{0,k}$),

$$\begin{aligned} s_{x,k}(t) = & \frac{A_{00}}{2} + \sum_{n=1}^{\infty} [A_{0n} \cos(n[\omega_{0,k}t + \theta_{0,k}]) + B_{0n} \sin(\cdot)] \\ & + \sum_{m=1}^{\infty} [A_{m0} \cos(m[\omega_{c,k}t + \theta_{c,k}]) + B_{m0} \sin(\cdot)] \\ & + \sum_{m=1}^{\infty} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} [A_{mn} \cos(m[\omega_{c,k}t + \theta_{c,k}] \\ & + n[\omega_{0,k}t + \theta_{0,k}]) + B_{mn} \sin(\cdot)], \end{aligned} \quad (9)$$

the cell's converter-side current follows from convolving each phase's switching-function spectrum with its current spectrum in the frequency domain, $I_{\text{conv},x,k} = \frac{1}{2}(S_{x,k} \otimes I_{x,k})$, and summing over the three phases [43]. For a single cell, this shows every carrier harmonic m contributes sidebands at $n = \pm 6p - 3$ (m odd) or $n = \pm 6p$ (m even), for integer p – richer than the single-tone picture of (8). With uniform interleaving $\theta_{c,k} = 2\pi k/N$, summing the N cells' converter currents shows that only carrier harmonics m that are themselves multiples of N survive ($m = pN$ for even N , $m = 2pN$ for odd N), each retaining the same $n = \pm 6p - 3/\pm 6p$ sideband structure [43]. This confirms, from the complete switched-current spectrum rather than a single-tone approximation, that the cancellation condition of (7) extends to the full family of sidebands around each carrier harmonic, not just its fundamental tone.

The same reference derives the resulting dc-side current directly from the SPB equivalent circuit already used in Sections III and VII – $L di_{\text{dc}}/dt = E - Ri_{\text{dc}} - \sum_k v_{\text{conv},k}$, $C_k dv_{\text{conv},k}/dt = i_{\text{dc}} - i_{\text{conv},k}$, the same governing equations underlying the coupled circuit simulation of Fig. 9 – giving a closed-form frequency-domain transfer function

$$I_{\text{dc}}(\omega) = \frac{\sum_{k=1}^N I_{\text{conv},k}(\omega)}{N - \omega^2 LC + j\omega CR} \quad (10)$$

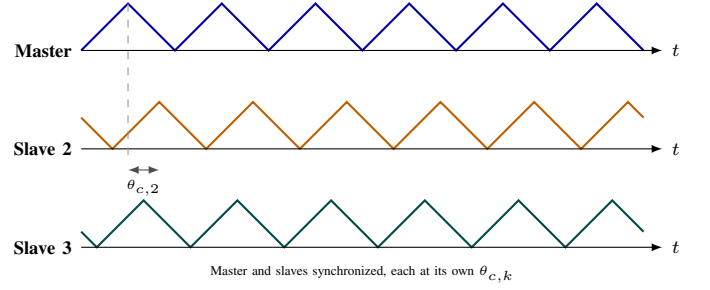


Fig. 8. Conceptual illustration, for a representative $N = 3$ example, of a master cell and two slave cells synchronized by the distributed master/slave mechanism of [48], each slave offset from the master by its own carrier-phase offset $\theta_{c,k}$ (7).

between the summed converter-current spectrum and the resulting dc-bus current [43].

In a centralized controller, $\phi_{c,k}$ in (7) is simply an offset applied to a single shared carrier timer, but SPB cells are physically distributed and do not share a clock. In practice, the phase relationship established by (7) is maintained through inter-cell communication: a designated master cell (or central controller) broadcasts a periodic synchronization pulse over the serial link, and each slave cell uses this pulse to align and offset its own local carrier timer, as demonstrated for the SPB converter in [48] and illustrated conceptually in Fig. 8 for a master cell and two slave cells.

The optimum carrier displacement obtained from (7)–(8) is not universal: it depends on cell count, modulation index, power factor, winding displacement, and the specific quantity being minimized. In particular, cancellation of one dc-link harmonic does not guarantee minimum machine harmonic loss or minimum common-mode current. Carrier coordination should therefore be viewed as one design variable within a larger converter-machine modulation problem, rather than as a self-contained optimization.

The resulting effect on the aggregate dc-link ripple current is illustrated by simulation in Fig. 9 for the same $N = 3$ case.

This simulated reduction already accounts for one lumped bus/wire inductance between the battery and the series stack, but still treats each cell's own local interconnect to that shared bus as ideal. In a physically distributed SPB implementation, each cell's contribution instead reaches the shared dc path through its own additional bus-bar and connector loop, and the parasitic inductance of that loop couples back into the ripple-cancellation mechanism itself: the 2015 IEEE TIA study by Wang, Li, and Han showed that in a series-connected GaN IMMD, the interleaved structure required careful capacitor selection and tight layout control because parasitic inductance and switching-generated CMV limited the achievable dc-link ripple reduction [39]. The same practical limitation is visible in the earlier interleaved series-connected-drive characterization, where parasitic loop inductance prevented full ripple cancellation and, at switching frequencies around 100 kHz, could make the interleaved current ripple *larger* than the non-interleaved case [84]. The simulated reduction of Fig. 9, which already accounts for local capacitor filtering but not bus-bar/connector parasitic inductance, should therefore still be read as an upper

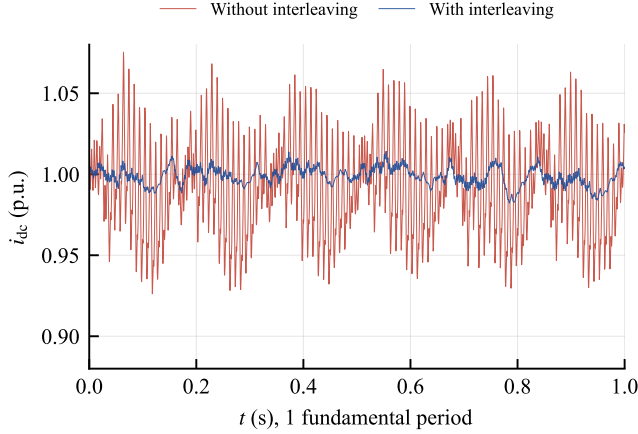


Fig. 9. Coupled circuit simulation of the shared series dc-link current $i_{dc}(t)$ for $N = 3$ SPB cells ($M = 0.85$, $\cos \varphi \approx 0.94$, $f_1 = 1$ Hz, $f_{sw} = 100$ Hz), without (red) and with (blue) carrier interleaving applied *between the three submodules* (each cell's own carrier phase-shifted, not the fixed 120° a/b/c reference separation within a cell), using the same mechanism as (7). Each cell's own naturally-sampled SPWM switching function drives a local converter-side current $i_{conv,k}(t)$; because the N cells are connected in series, one common current $i_{dc}(t)$ flows through the shared bus/wire inductance from the battery, and each cell's local dc-link capacitor C_k (Section III) supplies exactly the difference $i_{dc} - i_{conv,k}$, per $C_k dV_k/dt = i_{dc} - i_{conv,k}$ and $L di_{dc}/dt = V_{batt} - Ri_{dc} - \sum_k V_k$, solved self-consistently (shown after transients settle). Both traces share the same ≈ 1 p.u. dc average (set by the converters' average power draw, independent of interleaving, consistent with (10)); interleaving reduces the peak-to-peak ripple by roughly 4–5 $\times$. The corresponding harmonic content is characterized analytically, not empirically, by the double-Fourier survival condition derived above ((9)–(10)): only carrier harmonics that are themselves multiples of N survive interleaving, each still carrying its own $n = \pm 6p - 3/\pm 6p$ sideband family. This is a computed circuit-simulation result, not measured hardware data.

bound on what carrier interleaving can achieve in a physically distributed implementation, with the achievable benefit set by physical layout, capacitor selection, and switching frequency rather than by carrier phasing alone.

B. Machine Harmonics, Voltage Utilization, and Switching Loss

The multiphase machine identified in Section IV makes the harmonic problem substantially more involved than in a conventional three-phase drive. Nonfundamental voltage components can excite additional harmonic subspaces with low impedance, producing copper loss and torque ripple even when fundamental current control is fully satisfactory. Recent integrated-modular-drive studies accordingly move toward interleaved and discontinuous PWM techniques tailored to this multiphase current-ripple problem [85]–[87].

At high speed, modulation must also maximize voltage utilization, because each winding group is supplied only from its local cell voltage $V_{dc,k}$, as constrained by (6). Optimized vector selection for dual-three-phase traction machines has demonstrated the possibility of extending the high-modulation region while simultaneously suppressing unwanted subspace currents [64]. This objective becomes increasingly important as N grows and $V_{dc,k}$ decreases accordingly.

Switching loss introduces an opposing objective. DPWM and hybrid strategies reduce the number of transitions

TABLE III
SPB MODULATION DEGREES OF FREEDOM GROUPED BY PRINCIPAL SYSTEM-LEVEL EFFECT.

Effect	Degrees of freedom
Ripple / loss shaping	Carrier phase [47], [51], [85]; switching sequence [83], [94]; clamping/DPWM [88], [89]
CMV / balancing authority	Zero-sequence & vector redundancy [91], [92]; differential cell commands [54], [90]

by clamping selected phases or by redistributing high-frequency switching among converter modules. Related open-end-winding (OEW) work demonstrates that coordinated modulation can reduce switching loss and unwanted zero-sequence current simultaneously [88], [89]. Although the dc architecture differs from SPB, the underlying principle of distributing switching activity among several inverter modules is directly transferable.

C. Coupling to Balancing and Common-Mode Objectives

Modulation also interacts directly with the dc-link balancing problem introduced in Section II and formalized in Section VII: differential winding-group power must be realizable through modulation without violating local voltage and current limits [54], [90]. Common-duty-ratio operation further shows that propagation delay and synchronization among cells can influence the expected voltage-sharing behavior [55]. Common-mode-oriented multiphase modulation adds a further objective, using redundant voltage vectors or coordinated zero-sequence terms to reduce frame excitation [91], [92] — a concern reinforced by direct measurements of electromagnetic interference in physically integrated motor-converter assemblies [93].

Table III groups these degrees of freedom by principal system-level effect.

The key SPB modulation question is therefore not which conventional PWM method performs best in isolation, but how switching decisions should be coordinated across cells to jointly manage the objectives above over the full operating map; a future SPB-specific multiobjective study should report system-level converter-machine metrics rather than a single THD or switching-loss figure of merit.

VI. COMMON-MODE VOLTAGE EFFECTS

In any converter-fed ac machine, the non-zero average of the three phase-leg pole voltages produced by PWM switching gives rise to a common-mode voltage (CMV) that excites parasitic capacitive paths between winding, frame, shaft, and bearings. This mechanism is well characterized for a converter with a single machine port whose voltage scaling is produced internally to the converter, such as the flying-capacitor multilevel converter, for which an analytical CM-EMI model referenced to one dc-link potential is available [95]. SPB does not fit this picture. Because its N converter cells are connected in series on the dc side (Section III) while each cell feeds an electrically separate winding group, the local dc rails of cell k are displaced from the overall dc-bus reference by an amount that depends on the cell's position in the series stack. The CMV problem in SPB is therefore not N independent copies

of the single-inverter case: it is N common-mode sources referenced to N different absolute potentials, coupled through a shared machine frame, shaft, and cooling structure.

A. Series-Stack Origin of Common-Mode Excitation

Taking cell 1's local negative dc rail as the reference and stacking the remaining cells so that cell k 's negative rail coincides with cell $(k-1)$'s positive rail, the absolute offset of cell k is

$$V_{-,k} = \sum_{j=1}^{k-1} V_{dc,j} \approx (k-1) \frac{V_{dc}}{N}, \quad (11)$$

using the voltage-partitioning relation (1). The common-mode potential seen by winding group k relative to this fixed reference is then

$$v_{cm,k}^{abs}(t) = V_{-,k} + \frac{1}{3}(v_{aN_k} + v_{bN_k} + v_{cN_k}), \quad (12)$$

where $v_{aN_k}, v_{bN_k}, v_{cN_k}$ are the pole voltages of cell k referenced to its own local negative rail N_k . The second term is the conventional switching-induced CMV already addressed by zero-sequence injection and vector-redundancy modulation (Section V); the first term, $V_{-,k}$, is specific to SPB and is set by the cell's position in the stack and by the instantaneous voltage-sharing state discussed in Section VII, not by modulation alone.

Fig. 10 illustrates the resulting structure: increasing absolute potential offset with stack position, together with representative common-mode current paths coupling each winding group to the shared machine frame and chassis ground through parasitic capacitance. Because the converter cells and their winding groups are physically close together in the integrated implementations discussed in Section IV, these parasitic paths are short and the electromagnetic coupling between cells is direct rather than mediated by long cabling. Physically integrated motor-drive assemblies show that common-mode current can resonate through exactly this kind of parasitic loop, and that the resulting resonance is not adequately predicted by a single-source CM model [93], [96]. Extending such modeling from a single CM source to a multi-port structure with N distinct dc offsets is, to the authors' knowledge, not yet available for SPB specifically – the gap left open by the single-port flying-capacitor comparison drawn above [95].

For the special case $N = 2$, an SPB-specific mitigation option exists that is structurally different from the stack-offset mechanism above. If cell 2 is driven with the logical inverse of cell 1's gate commands, and cell 2's winding set is wired with reversed winding sense so that both cells still generate torque in the same direction, the two cells' individually switched common-mode voltages become approximately equal and opposite, $v_{cm,1}(t) \approx -v_{cm,2}(t)$, and the total common-mode voltage seen by the machine,

$$v_{cm,tot}(t) = \frac{v_{cm,1}(t) + v_{cm,2}(t)}{2}, \quad (13)$$

cancels toward zero instead of adding constructively. This balanced-cancellation principle was first demonstrated for a

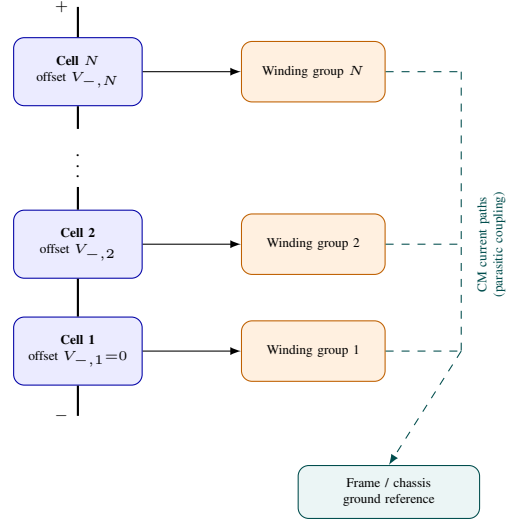


Fig. 10. Simplified SPB series-stack structure showing the offset absolute potential of each cell/winding group and representative common-mode (CM) current paths (dashed) coupling each winding group to the shared machine frame and chassis ground through parasitic capacitance. Cell k 's local dc rails are displaced from the reference of cell 1 by $V_{-,k} = \sum_{j=1}^{k-1} V_{dc,j}$, so each winding group's common-mode excitation combines this stack offset with its own local switching waveform.

dual-winding-set machine driven by a balanced inverter pair on a single, non-series-stacked machine [97], and was later applied to a genuine two-cell series-stacked (2L-SSC) SPB inverter [57]; as already noted for interleaved GaN IMMDS, the achievable CMV reduction is likewise ultimately limited by layout parasitics and local dc-link capacitor design [39]. This cancellation assumes perfectly matched, simultaneous switching transitions between the two cells; a propagation-delay mismatch between the two cells' gate signals (e.g., from PCB layout or driver differences) reintroduces a residual common-mode spike, of duration equal to the delay, at every misaligned switching transition, and unequal power sharing between the two cells during dc-link balancing (Section VII) similarly degrades the cancellation, both shown quantitatively in [57], [97]. This pairwise-cancellation mechanism is specific to $N = 2$ (or to a stack of matched cell pairs) and does not generalize to arbitrary N , so it should be read as a valuable special-case mitigation option rather than a general solution to the stack-offset common-mode problem discussed above. Fig. 11 simulates this cancellation mechanism directly from switching-function waveforms, including its effect on the resulting bearing/ground current.

B. Physical Consequences and Mitigation

CMV-driven bearing current is a documented failure mechanism in inverter-fed machines generally, arising from capacitive coupling through the air gap and from discharge events once the bearing lubricant film breaks down [98], and it can be identified with dedicated non-intrusive current-measurement techniques [99]. In SPB, this mechanism is present at each of the N winding groups individually, and because (11)–(12) gives each winding group a different absolute CM reference, the resulting bearing- and shaft-voltage stress is not necessarily

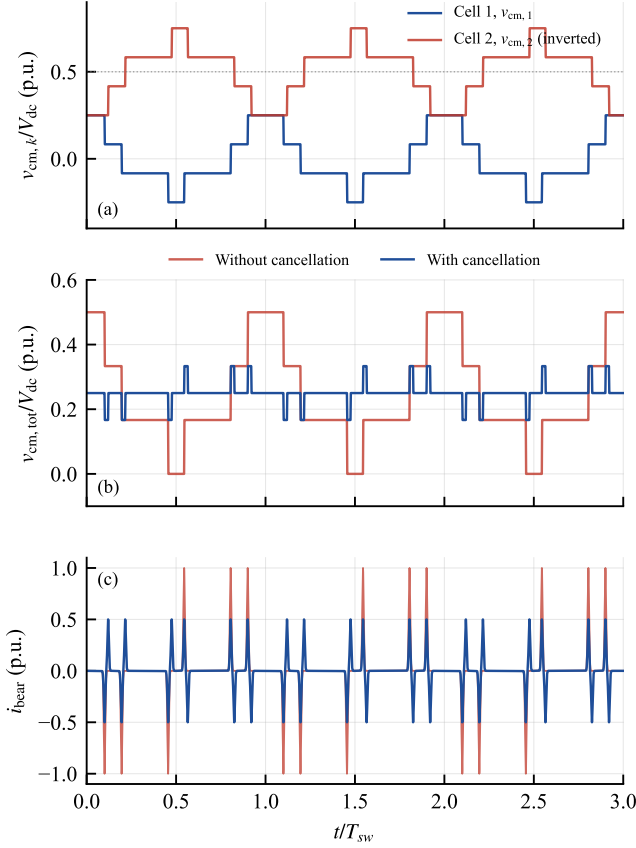


Fig. 11. Switching-function simulation of $N = 2$ SPB common-mode cancellation ($M = 0.85$, $\cos \varphi \approx 0.94$), all quantities in per unit. (a) Each cell's own *absolute* common-mode voltage $v_{cm,k}^{abs} = V_{-,k} + [(S_a + S_b + S_c)/3 - \frac{1}{2}]/N$, combining the stack offset $V_{-,k}$ (11) (dashed line marks $V_{-,2} = V_{dc}/N = 0.5$) with a six-step local switching term from the three independent phase-leg transitions; cell 2 uses the logically inverted gate commands of cell 1 plus a small inter-cell propagation delay $t_d = 0.02 T_{sw}$. (b) Total common-mode voltage $v_{cm,tot}(t)$ (13): without cancellation the local terms add, swinging fully around the same stack-offset average; with cancellation the local terms cancel, leaving only narrow delay-induced residual spikes on top of that same average – both settle to $\approx 0.25 = \text{mean}(V_{-,1}, V_{-,2})$, confirming that pairwise cancellation removes the switching ripple but not the structural stack offset itself. (c) Resulting bearing/ground current through a capacitive coupling path, $i_{bear} = C dv_{cm,tot}/dt$, normalized to the without-cancellation peak: verified numerically that the constant stack offset in (b) contributes zero current (a constant has zero derivative) even though it dominates the voltage, so the current with cancellation is driven entirely by the delay-induced spikes. This is a computed simulation result, not measured hardware data.

uniform across the stack. Whether it is dominated by the highest-offset cell, by the combined contribution of all cells, or by resonant interaction between cells is an open question that the direct SPB literature has not yet addressed quantitatively. Shaft-voltage buildup and associated common-mode noise are also targeted directly by cancellation-based circuits developed for conventional drives [100]; such circuits are a plausible mitigation candidate for SPB but have not been demonstrated for a series-stacked, multi-port machine.

The same switching events that generate CMV also produce fast dv/dt transitions at the winding terminals, which stress turn-to-turn and phase-to-frame insulation independently of the CM current path itself. WBG devices, already motivated

for SPB by their favorable scaling at reduced blocking voltage (Section VIII), shorten rise time and increase overshoot, and have been shown to accelerate insulation aging and partial-discharge activity in hairpin and other traction-machine windings [101]–[103]; frequency-response and other diagnostic methods have been developed to assess the resulting insulation condition [104], within the broader set of insulation technologies available for transport-electrification machines [105]. In SPB, this dv/dt stress is superimposed on the stack-position-dependent offset of (11), so insulation coordination must account for both the local switching transition and the cell's absolute position in the series stack, consistent with the insulation challenge already identified qualitatively in Section IV.

Mitigation therefore has to act at more than one level. Active EMI filtering can attenuate common-mode current at the converter terminals without relying on machine or bearing hardware [106], and cancellation circuits can target shaft voltage directly [100]. At the modulation level, Section V already showed that zero-sequence injection and redundant-vector selection can reduce CMV excitation [91], [92]. However, the same section also showed that the carrier-phase displacement used to interleave switching harmonics for dc-link and machine ripple reduction, (7)–(8), is chosen to cancel a selected ripple harmonic and is not simultaneously optimized for common-mode excitation. Because SPB's CM problem additionally depends on the fixed stack offset $V_{-,k}$, which carrier phasing cannot change, an interleaving choice that minimizes dc-link or stator current ripple need not minimize aggregate common-mode excitation across the stack, and vice versa. This is a genuine SPB-specific design tension: the multi-objective modulation problem identified in Section V should be extended, in future SPB-specific studies, to include common-mode and insulation-stress metrics evaluated across the full stack rather than at a single representative cell.

VII. DC-LINK DYNAMICS, STABILITY, AND VOLTAGE BALANCING

The shared dc current and independently controllable winding-group powers (Section III) make the local capacitors coupled energy-storage states whose voltages are not guaranteed to remain equal [45], [46]. An independent study of a series-connected multilevel motor drive reached the same conclusion, showing that under distributed closed-loop current control the series-connected module voltages are inherently unstable without dedicated balancing action [84].

For cell k ,

$$C_k V_{dc,k} \frac{dV_{dc,k}}{dt} = V_{dc,k} i_{dc} - p_{ac,k} - p_{loss,k}. \quad (14)$$

This equation captures the essential balancing mechanism: the dc input power depends directly on local cell voltage because the series current is common, whereas the output power depends on the corresponding machine winding group and its controller.

For analysis, define

$$\tilde{V}_k = V_{dc,k} - \frac{1}{N} \sum_{j=1}^N V_{dc,j}. \quad (15)$$

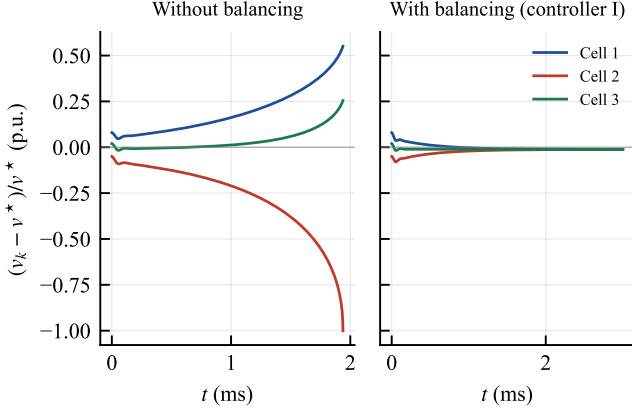


Fig. 12. Nonlinear simulation of SPB dc-link voltage-sharing dynamics for $N = 3$ cells, integrating the exact constant-power-load model of [46] (not its linearization) from the same initial per-cell voltage deviations (+8%, -5%, +2% of v^*) with and without their proportional balancing controller. Without balancing (left), the open-loop instability proven in [46] drives Cell 2's voltage toward collapse within about 2 ms while Cells 1 and 3 compensate; with balancing (right, same initial conditions, same time axis), all three cells converge back toward v^* on a comparable timescale. This is a computed simulation result using real experimental circuit parameters from the cited reference, not measured hardware data.

A simplified differential-mode model around a balanced point is

$$CV_0 \frac{d\tilde{V}_k}{dt} \approx I_{dc,0} \tilde{V}_k - \Delta P_k. \quad (16)$$

In motoring operation, $I_{dc,0} > 0$; a positive cell-voltage perturbation increases the dc input power of that cell and can reinforce the voltage deviation if the ac-side power remains unchanged. In regeneration the sign of this feedback reverses. This simplified interpretation is consistent with the detailed SPB stability analyses in [45], [46] and with recent work on related series-connected multiphase drives [107].

Reference [46] derives this instability from the full nonlinear constant-power-load model directly, without the small-signal approximation of (16): with $L_b di_{dc}/dt = E_b - R_b i_{dc} - \sum_k V_{dc,k}$ and $C_k dV_{dc,k}/dt = i_{dc} - P_k/V_{dc,k}$, the open-loop characteristic polynomial has $N - 1$ real, strictly positive eigenvalues in motoring operation ($P_k > 0$) for any $N > 1$, so the series stack is provably unstable without a balancing controller, regardless of parameter choice. Fig. 12 integrates this exact nonlinear model directly (not its linearization), using the real experimental parameters of [46] ($L_b = 8 \mu\text{H}$, $R_b = 0.35 \Omega$, $C = 100 \mu\text{F}$ per cell): without a balancing controller, a small initial voltage imbalance grows into a full collapse of one cell's voltage within milliseconds, while the proportional balancing law of [46] ($\Delta P_k = g'(V_{dc,k} - \bar{V}_{dc})$, their “controller alternative I”) drives the same initial imbalance back to zero on a comparable timescale.

The multiple winding groups provide a natural actuator for controlling internal capacitor energy. The cell power reference can be decomposed as

$$P_k^* = \frac{P^*}{N} + \Delta P_k, \quad \sum_{k=1}^N \Delta P_k = 0, \quad (17)$$

TABLE IV
PRINCIPAL SPB DC-LINK BALANCING APPROACHES AND THEIR MAIN CONSIDERATIONS.

Approach	Mechanism	Key consideration
Centralized	Differential current/power commands [45], [46]	Direct authority; global sensing and coordination
CDR	Common modulation, natural coupled dynamics [52], [55]	Low overhead; symmetry- and timing-sensitive
Distributed	Local agents, neighbour communication [53], [54], [56]	Scalable; requires communication and synchronization

so differential power changes the internal energy distribution without changing the requested total machine power. In practice, ΔP_k is limited by available phase-current margin, winding voltage, torque-ripple constraints, and thermal state. Balancing authority is therefore operating-point dependent.

Three main SPB approaches can be identified. Centralized active balancing uses measured capacitor voltages to generate differential current or power commands [45], [46]. Common-duty-ratio (CDR) control instead applies common modulation commands and can obtain stable voltage sharing from the natural converter-machine dynamics for the investigated induction-machine system [52]; practical timing mismatch remains important [55]. Distributed multi-agent control uses local measurements and neighbour communication and has been demonstrated for voltage balancing, power sharing, and reconfiguration [53], [54], [56], comparable to how local droop laws coordinate power sharing across modules in the modular speed-droop framework of Galassini *et al.* [108]. Table IV summarizes these three approaches and their principal considerations.

Fig. 13 compares the three approaches of Table IV directly, each drawn as the control block diagram for a representative cell k : a summing junction adds a per-cell correction $\Delta v_{d,k}, \Delta v_{q,k}$ ahead of the $dq \rightarrow \alpha\beta$ transform and PWM stage, following the structure used by [46] for their “controller alternative I”. The three panels differ only in what feeds that correction.

For automotive operation, balancing must remain effective through torque transients, field weakening, parameter mismatch, and module reconfiguration. Voltage-sensor offset, communication latency, propagation delay, and the available differential current/power margin can therefore determine the usable cell-voltage margin and, consequently, the semiconductor rating.

This underscores the section's central point: dc-link stability belongs to the complete converter-machine-control system, not the power stage alone.

VIII. SEMICONDUCTOR TECHNOLOGY AND CELL-COUNT SCALING

Semiconductor voltage scaling (Section I-B) and the voltage-versus-component-count tradeoff established in Section III are quantified here at the cell-count level. In an N -cell SPB, the dc-link voltage is distributed among the series-

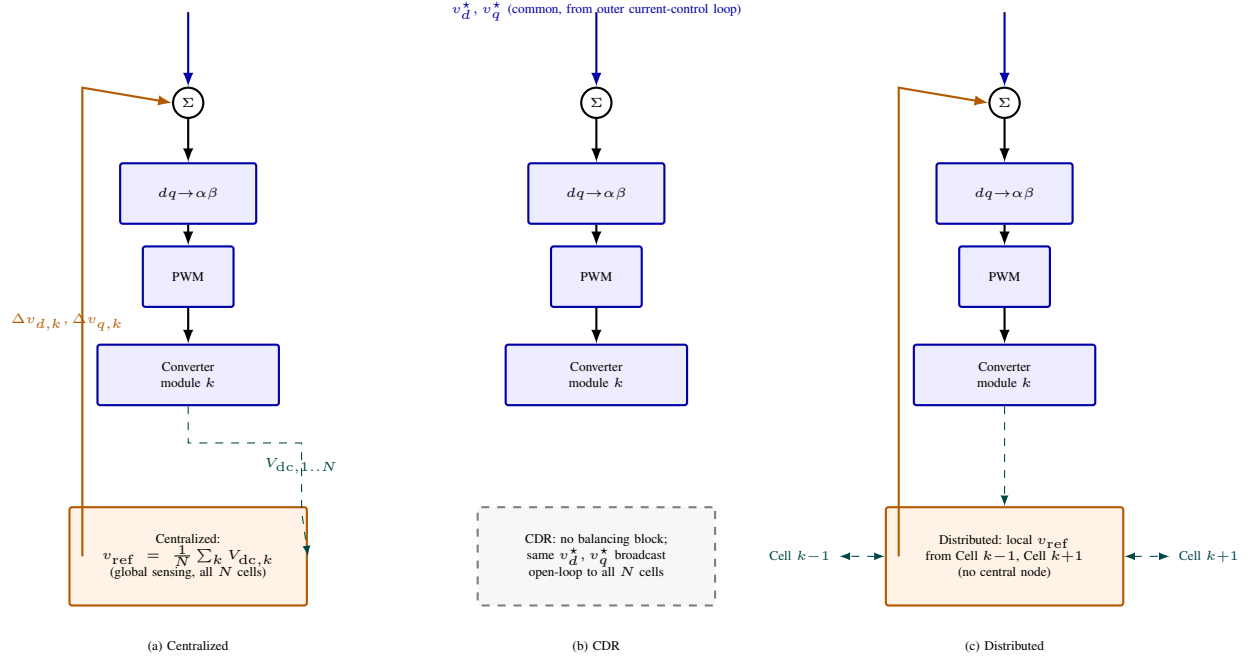


Fig. 13. Control block diagram for a representative cell k under each of the three SPB dc-link balancing approaches of Table IV. All three share the same forward path – a common current-control loop sets v_d^*, v_q^* for every cell, and a summing junction ahead of the $dq \rightarrow \alpha\beta$ /PWM stage admits a per-cell correction $\Delta v_{d,k}, \Delta v_{q,k}$ – following the structure used for the “controller alternative I” simulation of Fig. 12 [46]. (a) Centralized: one controller senses every cell’s $V_{dc,k}$ and computes $v_{ref} = \frac{1}{N} \sum_k V_{dc,k}$ [45], [46]. (b) CDR: the balancing block is absent entirely ($\Delta v_{d,k} = \Delta v_{q,k} = 0$); all cells receive the same open-loop command and balancing relies on the natural coupled converter–machine dynamics [52]. (c) Distributed: each cell computes its own local v_{ref} from its own voltage and neighbour-only measurements exchanged with Cell $k-1$ and Cell $k+1$, with no central node [53].

connected converter cells, allowing the use of lower-voltage semiconductor classes as N increases.

The device rating must be selected from the maximum expected blocking stress, not the nominal voltage division alone:

$$V_{rated} \geq k_{rel} \left(\frac{V_{dc,max}}{N} + \Delta V_{imb}^{max} + \Delta V_{sw}^{max} \right), \quad (18)$$

where ΔV_{imb}^{max} denotes the maximum cell-voltage imbalance, ΔV_{sw}^{max} the commutation overshoot, and k_{rel} the selected reliability margin. Cell count is therefore linked to discrete commercial voltage classes rather than to a continuous semiconductor-scaling law.

As introduced in Section I-B, SiC currently provides the most direct route for high-voltage traction, whereas GaN becomes increasingly attractive once the architecture reduces the per-device voltage enough to access lower-voltage classes with favorable conduction and switching characteristics [24], [36], [57].

The resulting discrete design transitions are summarized and visualized directly in Fig. 14. For a 1200-V dc link, $N = 2$ gives a nominal cell voltage of 600 V and therefore leaves limited transient margin for a 650-V device. Increasing the cell count to $N = 3$ reduces the nominal cell voltage to 400 V and provides a substantially more practical operating region for the 650-V class. Further increases in N give access to progressively lower-voltage device families, as labeled directly on Fig. 14. The potential semiconductor benefit is therefore nonlinear because a small change in N can open a new commercial voltage class, whereas the active-switch count

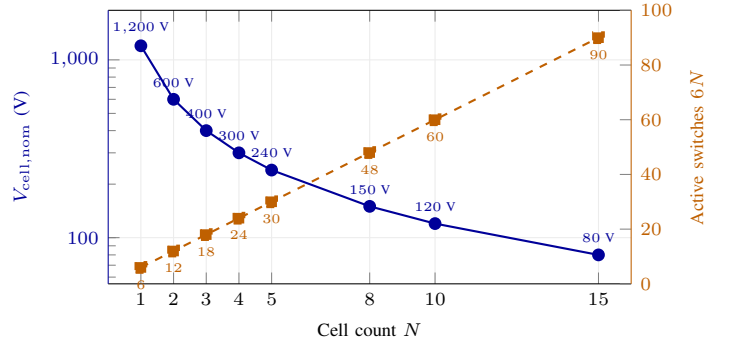


Fig. 14. Cell-count scaling for a 1200-V dc link: nominal cell voltage $V_{cell,nom} = V_{dc}/N$ (solid markers, left log axis) and active-switch count $6N$ (dashed markers, right linear axis) versus cell count N ; every marker is labeled directly with its exact value ($N = 1$: 1200 V, 6; $N = 2$: 600 V, 12; $N = 3$: 400 V, 18; $N = 4$: 300 V, 24; $N = 5$: 240 V, 30; $N = 8$: 150 V, 48; $N = 10$: 120 V, 60; $N = 15$: 80 V, 90). Increasing N opens progressively lower commercial semiconductor voltage classes (e.g., 1700–2000-V SiC at $N = 1$, 650-V GaN/SiC at $N = 3$, 100-V GaN/MOSFET class at $N = 15$) while the active-switch count grows linearly as $6N$.

grows approximately linearly, as plotted directly from these design points in Fig. 14.

A. Commercial-Device Screening

Access to a lower voltage class does not by itself imply a superior converter: the comparison must be based on commercial devices and their conduction, switching, current, thermal, and packaging constraints.

For nonlinear output capacitance, the stored energy

$$E_{\text{oss}}(V) = \int_0^V v C_{\text{oss}}(v) dv \quad (19)$$

is more informative than a single tabulated C_{oss} value. Depending on the commutation process, some or all of this stored energy can contribute to switching loss. Similarly, simple screening metrics such as $R_{\text{DS(on)}}C_{\text{oss}}$ or $R_{\text{DS(on)}}Q_g$ are useful for identifying candidate devices, but are insufficient for the final architecture comparison.

The principal device quantities required for SPB cell-count screening, and their primary role in the comparison, are: $R_{\text{DS(on)}}(T)$, which sets the die-area or parallel-device requirement; $E_{\text{oss}}(V)$, which sets the large-signal switching penalty; $E_{\text{on/off}}(V, I, T)$, which forms the operating-point switching-loss model; gate charge Q_g and Q_{gd} , which govern driver sizing and auxiliary loss; continuous current capability I_{cont} and thermal resistance R_{θ} , which govern parallelization and cooling requirements; and package or loop inductance L_{σ} , which sets the usable blocking-voltage margin against di/dt -induced overshoot.

Current capability is particularly important because machine phase current does not generally decrease in proportion to $1/N$ (Section III). If the derated current capability of one device is $I_{\text{dev,der}}$, the required number of parallel devices can be estimated as

$$n_p = \left\lceil \frac{I_{\text{sw,max}}}{I_{\text{dev,der}}} \right\rceil, \quad N_{\text{sw,tot}} = 6Nn_p. \quad (20)$$

Thus, a lower-voltage device can offer an attractive device-level FOM while still requiring substantial die area, package count, and gate-drive/thermal interfaces. Recent high-current 650-V GaN modules illustrate the importance of low-inductance packaging and integrated gate driving when GaN is scaled toward traction-current levels [109], [110].

Fast switching further couples device selection to layout. The commutation overshoot can be approximated by

$$\Delta V_{\text{sw}} \approx L_{\sigma} \frac{di}{dt}, \quad (21)$$

so the benefit of reduced nominal cell voltage can be lost if package and interconnect inductance consume a substantial fraction of the available blocking margin. This is particularly relevant for GaN, where high di/dt places strong requirements on the commutation loop, local decoupling, and gate-drive layout [110].

GaN device selection also requires consideration of dynamic $R_{\text{DS(on)}}$, reverse conduction, gate reliability, short-circuit capability, and thermal performance [111], [112]. These effects do not negate the benefit of lower blocking voltage, but they prevent room-temperature static $R_{\text{DS(on)}}$ from being used as the sole criterion for cell-count selection.

B. Loss Scaling With Cell Count

For one device, the dominant semiconductor losses can be represented approximately as

$$P_{\text{cond}} \approx I_{\text{rms,sw}}^2 R_{\text{DS(on)}}(T_j), \quad (22)$$

and

$$P_{\text{sw}} \approx f_{\text{sw}} [E_{\text{on}}(V, I, T_j) + E_{\text{off}}(V, I, T_j)]. \quad (23)$$

At converter level,

$$P_{\text{semi}}(N) = \sum_{k=1}^N \sum_{j=1}^{6n_p(N)} (P_{\text{cond},k,j} + P_{\text{sw},k,j} + P_{\text{dt},k,j}) + P_{\text{gate}}(N). \quad (24)$$

The dependence on N is therefore not linear. The number of active devices increases with cell count, whereas $R_{\text{DS(on)}}$, switching energy, thermal impedance, and parallel-device requirement can change abruptly when the design enters a different semiconductor voltage class.

Consequently, SiC and GaN should be regarded as different regions of the SPB design space rather than as a fixed technology ranking. Low-cell-count architectures can favor high-voltage SiC because of its mature voltage and current capability, whereas increasing N can make lower-voltage GaN attractive if the improvement in conduction, switching, packaging, or passive-component performance compensates for the additional converter modules.

Resolving the cell-count question therefore requires reading Fig. 14 jointly with the screening quantities above and the machine requirements of Section IV, rather than from any single figure of merit.

IX. FAULT TOLERANCE AND RECONFIGURATION

SPB modularity provides structural redundancy because the converter and machine are divided into repeated converter-winding modules. However, fault tolerance is not automatic: because the dc ports are connected in series, a failed cell can disturb or interrupt the complete dc current path, a structural challenge demonstrated for SPB and for other series-connected modular converters alike [113]–[115]. Early SPB work demonstrated post-fault operation under semiconductor faults, establishing that repeated hardware must be accompanied by appropriate safe states and control action [113]. At the machine level, a submodule short-circuit fault in a modular converter feeding fractional-slot concentrated-winding PM machines was shown to drive a fault current above rated current because of the strong NdFeB rotor magnetization, risking machine overheating; reducing rotor magnetization limits this fault current but also narrows the constant-power-speed range, a design tradeoff specific to fault-tolerant FSCW PM machines [114]. Table V reports the measured post-fault short-circuit characteristics for the SPB machine designs in that study: with NdFeB magnets the short-circuit current exceeds the healthy-phase current, whereas with ferrite magnets it stays below it, at the cost of the short-circuit current becoming visibly non-sinusoidal due to magnetic coupling between the short-circuited winding and the adjacent healthy windings – a concrete illustration that PM material selection is itself a fault-tolerance design variable, not only a torque-density one. Fault handling in other series-connected modular converters further underscores that mitigation requires the control system to be explicitly designed for module isolation [115].

The machine provides a second layer of redundancy. Healthy winding groups can potentially redistribute current

TABLE V
REPORTED POST-FAULT SHORT-CIRCUIT CHARACTERISTICS FOR SPB
MACHINE DESIGNS [114]

Machine	PM type	$i_{a,sc}$ (A _{rms})	$i_{b,sc}$ (A _{rms})	$i_{c,sc}$ (A _{rms})	T (N·m)
1	NdFeB	119.8	117.7	110.6	106
2	Ferrite	66.4	27.5	57.0	105
3	Ferrite	60.6	34.3	56.8	103

$i_{x,sc}$: rms short-circuit current in phase x of the faulted submodule's winding group after a converter submodule short-circuit; T : resulting steady-state electromagnetic torque. Machine 1's healthy-phase current after redistribution was 74.2 A_{rms}, below all three of its short-circuit currents; Machines 2–3's healthy-phase currents (116.5/115.6 A_{rms}) exceed their short-circuit currents, consistent with the NdFeB-vs-ferrite contrast discussed in the text.

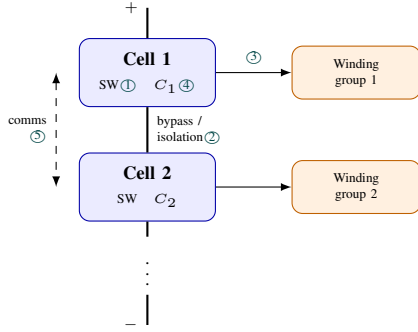


Fig. 15. Simplified SPB converter-machine structure with fault locations ①–⑤ annotated, corresponding in order to the rows of Table VI.

after phase or converter faults, as demonstrated in recent multiphase and modular-drive studies [62], [116], [117]. In SPB, however, this machine-side redundancy is useful only if the failed converter cell can be isolated or bypassed without breaking the series dc path.

Building on the online reconfiguration capability introduced in Section II [56], after a faulted module is isolated the remaining active set $\mathcal{A}$ must satisfy

$$P^* = \sum_{k \in \mathcal{A}} P_k^*, \quad (25)$$

with torque derating if current, voltage, or thermal limits prevent full redistribution. Reconfiguration can therefore provide both fault isolation and an operating-range degree of freedom because the number of active cells changes the voltage available to the remaining winding groups. Isolating a cell also raises the voltage that each surviving cell must share of the fixed dc-link voltage V_{dc} ((1), now divided among $|\mathcal{A}| < N$ active cells), which is an additional reason current or power derating in (25) may be needed after a fault, as illustrated in Fig. 16. Fig. 15 maps these fault locations onto the SPB structure.

Reliability remains ambiguous because increasing cell count provides more potential degraded states but also more components that can fail. System or mission reliability is therefore more meaningful than component count alone. Markov-chain analysis has been applied to related fault-tolerant IMMDS to distinguish healthy, degraded, and failed states [119], while integrated distributed traction drives provide additional evi-

TABLE VI
SPB FAULT LOCATIONS AND REQUIRED SYSTEM RESPONSE

Fault	SPB consequence	Required response
Switch / bridge leg	Cell cannot synthesize commanded voltage	Fast safe state and protection [113]
Converter module	Series dc path can be interrupted	Bypass/isolation and power redistribution [56]
Winding / phase	Reduced and asymmetric torque capability	Multiphase post-fault current control [62], [117]
Capacitor / sensor*	Loss of voltage-sharing information or margin	Detection, derating, isolation or shutdown
Communication / controller	Loss of distributed coordination	Local fallback and deterministic safe state [118]

*No SPB-specific study directly reports capacitor or sensor fault handling; this row reflects the authors' system-level inference from the general modular-converter requirement to detect loss of cell-voltage sensing or local dc-link capacitance and to derate or isolate the affected cell accordingly.

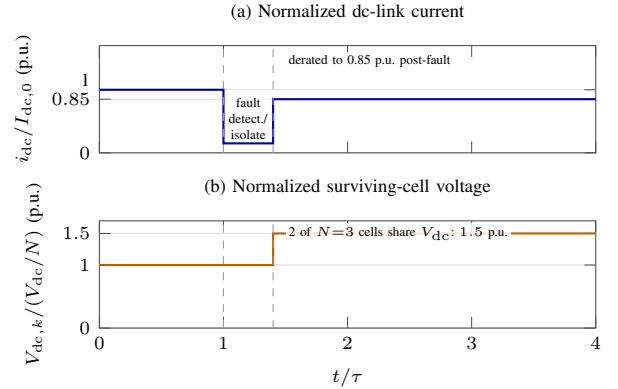


Fig. 16. Illustrative, analytically derived example of the post-fault current derating and remaining-cell voltage step-up for a representative $N = 3 \rightarrow 2$ case (one of three cells isolated after a fault); this is not measured or simulated data. (a) Normalized dc-link current, derated post-fault per (25). (b) Normalized remaining-cell voltage, stepping from 1 p.u. to $N/(N-1) = 1.5$ p.u. (1) once two cells must share V_{dc} . Both panels share the same time axis.

dence that fault tolerance must include both the machine and the local converter architecture [70], [71].

The direct SPB literature therefore demonstrates the feasibility of post-fault operation and online reconfiguration, but not yet automotive-level fault coverage. Detection and isolation time, short-circuit energy, capacitor and sensor faults, communication loss, post-fault thermal loading, diagnostic coverage, and deterministic functional-safety transitions remain important validation topics.

X. RESEARCH GAPS AND FUTURE DIRECTIONS

SPB operation, modulation, balancing, and modular control are established, but the evidence needed for automotive architecture selection remains incomplete. The key gaps are not additional proof-of-concept control schemes but high-power validation and system-level comparisons under common boundary conditions.

TABLE VII
SUGGESTED VALIDATION LADDER FOR AUTOMOTIVE SPB RESEARCH

Stage	Primary purpose	Minimum evidence
Cell / double-pulse	Device and commutation feasibility	Overshoot, switching energy, thermal impedance, protection
Multi-cell bench	Balancing and modulation	Cell-voltage spread, ripple, timing mismatch, transient stability
Dynamometer	Converter-machine interaction	Torque-speed map, efficiency, harmonics, CM/EMI, thermal limits
Mission / vehicle-relevant	Architecture value	Drive-cycle energy, cooling, fault derating, reliability and packaging

A. High-Power Validation and Standardized Benchmarking

Building on the laboratory-scale validation status noted in Section II, future prototypes should target several-hundred-kilowatt scale with realistic dc voltage, thermal, switching-overshoot, and EMI constraints, among others, benchmarked directly against a two- or three-level reference drive under the same machine requirements, semiconductor generation, cooling boundary, and mission. The scale of benefit that motivates this effort is not merely hypothetical: a segmented, interleaved traction-drive system operating at 55 kW has reported a 55–89% reduction in dc-bus capacitor ripple current relative to a conventional single-bridge drive [120], an order of magnitude above the power level of most direct SPB studies and a useful automotive-scale reference point for the interleaving benefit discussed in Section V.

Standardized reporting is also needed: published studies vary in power level, cooling, filtering, and semiconductor generation, and comparisons can shift materially with the chosen switching-energy interpolation, temperature correction, and current-derating assumptions. Reference operating points, efficiency maps, measured parasitics, and open device/loss-model data would improve reproducibility as semiconductor technology evolves.

A practical development path should also separate the questions that can be answered at different hardware scales; Table VII proposes a validation ladder so that a low-power proof of one control function is not read as evidence of automotive readiness.

A community benchmark defining one passenger-car and one heavy-duty traction case – common dc-voltage range, power, torque-speed envelope, coolant temperature, modulation limits, and EMC boundary, with a VECTO-derived mission for heavy-duty work – would let competing architectures be compared under comparable electrical and thermal duty without forcing a single reference machine. Reporting should extend the same way: beyond peak efficiency, prototypes should state cell-voltage spread, capacitor ripple current, common-mode current, and post-fault capability, among other operating-point metrics, linking the modulation, balancing, semiconductor, and reliability literatures into transferable evidence.

B. Cell Count, Reliability, and Cost Co-Design

The optimum cell count N remains the central unresolved design question: as established in Section III, the voltage-versus-component-count tradeoff shifts the optimum as SiC, GaN, packaging, and cooling evolve [3], [24]. A common-platform comparison of a few-cell 650-V GaN SPB against a many-cell low-voltage GaN SPB, using real commercial devices, would be valuable; the GaN comparison in [57] is a useful starting point but should be extended to automotive power with current parallelization, machine voltage/speed constraints, dc-link capacitance, cooling, and mission energy.

Converter, machine, and thermal models, among others, should be co-optimized rather than designed sequentially. Existing phase-count studies show machine design changes with SPB cell number [34]; the missing step is a complete automotive co-design using vehicle-level energy, volume, mass, cost, and reliability as a joint objective.

Reliability is part of that objective: future studies should combine semiconductor, capacitor, sensor, gate-driver, communication, and winding faults (Section IX; [56], [113]) in one state-based, Markov-style framework as used for related fault-tolerant IMMDS [119], keeping *component failure*, *module isolation*, and *loss of vehicle function* distinct. An N -cell drive may survive with $N - 1$ active modules, but its torque-speed envelope, module thermal loading, and dc-voltage margin all change, so reliability data should pair with a post-fault capability model – not a series reliability block diagram – to compare availability, limp-home power, and mission completion probability against efficiency.

Cost and sustainability assessment should use the same N -dependent framing: increasing N can reduce per-switch die cost but tends to increase isolated gate supplies, sensing channels, and assembly operations, among others, almost linearly, while a repeated converter-winding module can gain from common tooling and automated assembly – a balance likely to differ between high-volume passenger vehicles and lower-volume heavy-duty systems. The same accounting should separate production, operation, maintenance, and end-of-life impact over one vehicle mission and service life, since modular repair and longer service life can offset the added production impact of extra components.

Overall, SPB research must move from demonstrating that the architecture operates toward demonstrating where it is better: whether it gives a measurable vehicle-level advantage over mature two- and three-level architectures once machine design, thermal management, EMC, reliability, manufacturing, and cost are included.

XI. CONCLUSION

This paper reviewed stacked polyphase bridge (SPB) converters – series-stacked two-level bridges each feeding an electrically separated winding group – as a candidate architecture for future high-voltage traction drives, partitioning the dc-link voltage while introducing converter-machine modularity, distributed control, and fault-management opportunities.

The review also shows that the semiconductor voltage advantage cannot be separated from the complete traction

system. Increasing cell count adds converter, control, and insulation overhead, so SPB does not eliminate the complexity of high-voltage conversion but redistributes it between semiconductor technology, machine architecture, and control. WBG development makes this trade increasingly relevant, particularly where SPB can move a high-voltage traction system toward lower-voltage GaN devices; whether that approach is advantageous depends on whether the device-level gains compensate for the additional modular hardware and machine requirements.

The remaining step is application-level validation at representative power, dc voltage, thermal conditions, EMI constraints, and vehicle missions. SPB should ultimately be judged not as another multilevel converter, but as a modular traction-drive architecture that exchanges semiconductor blocking-voltage stress for converter-machine modularity. Its competitiveness depends on whether that exchange produces a measurable vehicle-level benefit.

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